Contradiction and Contrapositive

MTH 317H — Week 4

Name: _____

1 Warm-up

Let $x \in \mathbb{Z}$. Prove the implication

$$5x - 11 \text{ is even} \quad \Longrightarrow \quad x \text{ is odd.}$$

We will prove this statement in two ways: first by contradiction and then by contrapositive.

Preparation

Recall that an integer is *even* if it has the form $2k$, and it is *odd* if it has the form $2k+1$, for some $k \in \mathbb{Z}$. Every integer is exactly one of even or odd.

2 Proof by contradiction

Definition 2.1. To prove a statement P by *contradiction*, assume that its negation $\neg P$ is true and deduce an impossibility. The contradiction shows that $\neg P$ is false, so P must be true.

Preparation

Steps for a proof by contradiction.

1. State that the proof will use contradiction.
2. Assume the negation of the desired statement.
3. Reason from that assumption until a contradiction is reached.
4. Conclude that the negation is false, and therefore the original statement is true.

3 The warm-up by contradiction

Write

$$P: \quad 5x-11 \text{ is even}, \qquad Q: \quad x \text{ is odd}.$$

The warm-up asks us to prove $P \Rightarrow Q$.

Preparation

The negation of an implication is

$$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q.$$

Thus, to negate the warm-up statement, we assume both that $5x-11$ is even and that x is not odd. Since $x \in \mathbb{Z}$, "x is not odd" means that x is even.

Warning:

$$\neg(P \Rightarrow Q) \not\equiv (\neg P \Rightarrow \neg Q).$$

Proof

We proceed by contradiction. Assume that $5x-11$ is even and that x is even. Since x is even, we may write

$$x = 2k$$

for some $k \in \mathbb{Z}$. Then

$$\begin{aligned} 5x-11 &= 5(2k)-11 \\ &= 10k-11 \\ &= 10k-12+1 \\ &= 2(5k-6)+1. \end{aligned}$$

Because $k \in \mathbb{Z}$, we also have $5k-6 \in \mathbb{Z}$. Therefore the last expression shows that $5x-11$ is odd. This contradicts our assumption that $5x-11$ is even.
Hence x cannot be even. Since x is an integer, it follows that x is odd. Thus

$$5x-11 \text{ even} \quad \Longrightarrow \quad x \text{ odd}.$$

4 Proof by contrapositive

Definition 4.1. The *contrapositive* of the implication $P \Rightarrow Q$ is

$$\neg Q \Rightarrow \neg P.$$

An implication and its contrapositive are logically equivalent. Therefore, to prove $P \Rightarrow Q$, it is enough to prove $\neg Q \Rightarrow \neg P$.

For the warm-up statement,

$$\underbrace{5x - 11 \text{ even}}_{P} \implies \underbrace{x \text{ odd}}_{Q},$$

the contrapositive is

$$\underbrace{x \text{ even}}_{\neg Q} \implies \underbrace{5x - 11 \text{ odd}}_{\neg P}.$$

Proof

We prove the contrapositive. Suppose that x is even. Then

$$x = 2k$$

for some $k \in \mathbb{Z}$. It follows that

$$\begin{aligned} 5x - 11 &= 5(2k) - 11 \\ &= 10k - 11 \\ &= 10k - 12 + 1 \\ &= 2(5k - 6) + 1. \end{aligned}$$

Since $5k - 6 \in \mathbb{Z}$, this proves that $5x - 11$ is odd. We have proved the contrapositive, so the original implication follows:

$$5x - 11 \text{ even} \implies x \text{ odd}.$$

Preparation

The contrapositive method applies to implication statements. Before using it, identify the hypothesis P and conclusion Q, then prove $\neg Q \Rightarrow \neg P$.